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EEG AND SLOW CORTICAL POTENTIALS IN ANTICIPATION OF MENTAL TASKS WITH DIFFERENT HEMISPHERIC INVOLVEMENT

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Mental arithmetic is supposed to be processed by the left hemisphere, while pattern matching has been considered to be processed by the right hemisphere. These different mental tasks were presented to 16 right-handed male subjects. As a further variable task difficulty was varied for the two task categories. The task presentations were preceded by 6 sec warning stimuli which indicated the level of difficulty of the task. EEG and slow cortical potentials (SCPs) were recorded from temporal and precentral leads of both hemispheres. The reference was non-cephalic. During the anticipation interval beta-activity was enhanced above the hemisphere primarily involved in the particular task. SCPs failed to show a comparably clear pattern. A late negative component showed task-dependent asymmetries prior to easy tasks, but not prior to difficult tasks.

1. Introduction

Numerous studies showed that verbal, analytic, and serial information is processed mainly by the left hemisphere, whereas visual–spatial, 'gestalt' information is better processed by the right hemisphere (for reviews see Deegener, 1978; Gazzaniga, 1978; Kinsbourne, 1978). Psychophysiological studies have demonstrated electrocortical correlates of this lateralization. Thus, alpha power was found to be decreased in that hemisphere primarily involved in the task (Galin and Ornstein, 1972; McKee, Humphrey and McAdam, 1973; Schwartz, Davidson, Maer and Bromfield, 1974; Dumas and Morgan, 1975; Galin and Ellis, 1975; Ehrlichman and Wiener, 1980; Ornstein, Herron, Johnstone and Swencionis, 1979). According to Kinsbourne, ". . . expectancy of a . . . word or letter induces preparatory left hemispherric activation and so leads to better left vs right hemispheric processing accuracy and reaction time" (Kinsbourne, 1970, p. 194). One would assume that this asymmetrical preparation is indicated by electrophysiological responses associated with

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expectancy. The preparation of a task has been shown to be accompanied by a cortical negative shift during the warned foreperiod within the two-stimulus paradigm (Walter, Cooper, Aldridge, McCallum and Winter, 1964; Tecce, 1972; Gaillard, 1978; McCallum, 1979). However, results regarding the asymmetries of this contingent negative variation (CNV) are inconsistent. While Butler and Glass (1974) and Rebert and Lowe (1980) found the more pronounced CNV over the hemisphere which they assumed to be primarily involved in the task, McCarthy and Donchin (1978) showed task *in*dependent asummetries. Marsh and Thompson (1973) and Gazzaniga (1972) found no CNV asymmetries at all. This inconsistency of the results is probably not only due to methodological problems, but also to intervening variables such as task difficulty or task complexity, which are known to have effects on the amplitude of the CNV (Delse, Marsh and Thompson, 1972; Poon, Thompson, Williams and Marsh, 1974; Poon, Thompson and Marsh, 1976; Gaillard, 1978; Donchin, Kutas and McCarthy, 1977). Task complexity seems to involve different hemishperic processing strategies, although, results are contradictory. While Fontenot (1973), Dee and Fontenot (1973), and Umilta, Bagnara and Simons (1978) found a left visual field (i.e. right-hemispheric processing) superiority for high complex forms or complex geometrical figures, Hannay, Rogers and Durant (1976) demonstrated a right visual field (left hemisphere) superiority for both low and high complex forms but not for the intermediate ones. Willis, Wheatly and Mitchell (1979) and Ornstein, Herron, Johnstone and Swencionis (1979) showed alpha-activity asymmetry to be primarily dependent on the processing strategies which are necessary for task solution: the mental rotation of a spatial pattern induced more left hemispheric engagement, in contrast to the perception of this pattern without additional processing requirements (rotation). The present experiment was designed to investigate hemispheric asymmetries when subjects prepared for tasks that involved the right or the left hemisphere and varied in the level of difficulty. Slow cortial potentials (SCPs) and EEG power spectra were measured during the preparatory interval to elucidate differences in task preparation between the cerebral hemispheres.

2. Method

2.1. Subjects

Sixteen right-handed male student volunteers (age 18–30) were paid for their participation. Subjects were not included into the sample if they reported to be under current medication. It was assured that subjects had no left-handed family members. Subjects' handedness was assessed by the Edinburgh Handedness Inventory (Oldfield, 1971). The mean score of the present sample was 85.6 with a range from +50 to +100.

2.2. Design and procedure

Within a S1-S2 waiting paradigm subjects heard as a warning stimulus (S1) one of two discriminable acoustic stimuli (pure tones of 700 or 1200 Hz, 65 dB) for 6 sec each. S1-offset was followed by the presentation of either an arithmetic or a pattern matching task (S2). Each S2 was given for 2 sec. Wtithin one series of trials either an easy (e.g. 1 + 2 + 1 = 3) or a difficult (e.g. 4 + 7 – 5 = 8) arithmetic task was presented. Subjects had to evaluate whether the presented task was correct or incorrect by a right or left turn of a switch with the right (dominant) index finger. The response was required within 5 sec following S2-onset. In a second series of trials two patterns were presented simultaneously as S2. Subjects had to evaluate whether the two patterns were the same, or different.

In half of the trials tasks were more difficult in that one of the two patterns was rotated by 45° and patterns were more complex. The experimental session consisted of two series of 80 trials each. Within each series 40 difficult and 40 easy trials were presented in random order. The relationship between S1 frequency (low vs. high tone) and task difficulty was counterbalanced across subjects. The subjects' decision (right or wrong, same or different), represented by the direction of the lever switch, was counterbalanced across subjects too. Task difficulty and correctness of the presented solutions varied randomly within subjects. Response latencies exceeding 5 sec and false decisions produced a 90 dB noise burst. Intertrial intervals varied randomly between 14 and 26 sec around a mean of 20 sec. For eight subjects the arithmetic tasks were presented in the fisrt series, for the other eight the patterns were presented first. The two trial blocks were separated by a rest period of 5 min.

During the experiment subjects sat in a comfortable reclining chair in an electrically shielded, dimly lighted, and soundproof room. After electrode application subjects read task instructions. Four practice trials introduced the subject into the stimulus sequences and the response requirements. Practice trials were supervised by the experimenter.

2.3. Apparatus and physiological recordings

The visual S2 was presented on a video screen 2 m in front of subjects' eyes. At this distance the display subtended a visual angle of 6°. A PDP 8/e computer controlled the presentation of the stimuli and digitized the physiological data, which were stored on magnetic tape. Turning the lever to the left or to the right closed a microswitch. The time between S2-onset and the closure of the microswitch was measured to the nearest 1 msec by the PDP 8/e clock.

The EEG was recorded monopolarly from the precentral and the temporal lobes (C3, C4, T3, T4 according to the international 10–20 system) with a time constant of 30 sec. Grass flat silver disc electrodes, chlorided before each experimental session, were used. Grass paste EC 2 served as the conducting agent. Elec-

trode impedance was held below 5 kΩ. The reference electrodes were affixed to the manulium sternum and the seventh cervical. The influence of the ECG was minimized by a variable potentiometer of 20 kohm which connected the two reference sites. EEG activity was amplified by a modified Beckman A-C 9806 coupler with an upper cut-off frequency of 30 Hz (–3 dB) for the central positions and 10 Hz (–3 dB) for the temporal positions. Sampling rate was 100 Hz at C3, C4 and 20 Hz at T3, T4. A calibrator provided 50 μV square wave pulses to the EEG channel at the completion of each response interval. Vertical eye movements (VEOG) were recorded via Beckman electrodes affixed 1.5 cm above and below the left eye. Horizontal eye movements (HEOG) were recorded via electrodes affixed as near as possible to the lateral outer canthi of the left and the right eye. The EOG was sampled at a rate of 20 Hz.

2.4. Data reduction and analysis

The digitized data of the slow cortical potentials (SCPs) were collapsed to 0.2 sec points which were referred to a 1 sec prestimulus (S1) baseline. Trials in which the mean deviation exceeded 75 μV in the EEG, 100 μV in the VEOG, or 30 μV in the HEOG (referred to baseline), were excluded from the analysis. This resulted in th exclusion of 24.4% of the trials for all subjects. A principal component analysis (PCA) of the covariance matrix with varimax rotation (see Donchin and Heffley, 1979) was used to describe the components of the SCPs during the S1-interval. The number of components to be retained was set at the number of 'eigenvalues' equal to or greater than the trace of the covariance matrix divided by its dimension. An ordinary least square fit was performed to obtain the component (factor) scores as measure for the contribution of the component to the individual slow potential.

EEG power spectra of the precentral recordings were calculated by Fast Fourier Transformaion (FFT), using a Tukey–Hanning window, for two 2.56 sec intervals (from 1 to 3.56 and from 3.44 to 6.0 sec) within the S1-interval. Mean slow activity in the alpha range (8–12 Hz) and mean fast activity (12.4–20 Hz) were computed from the spectra normalized in the range 4–20 Hz.

Effects were statistically evaluated by ANOVAs with the within-subjects factors Difficulty (Diff, easy vs. difficult), Hemispheres (HE, left vs. right), and Task (arithmetic vs. patterns). The ANOVA for SCPs included the factor Topography (TO, central vs. temporal), and the analysis of the EEG power spectra included the additional factor Interval (INT, first vs. second part of the S1-interval).

3. Results

3.1. Response latencies

All subjects solved the easy tasks in a shorter time (1.98 sec mean latency from task presentation to lever switch) than the difficult ones (response after 2.30 sec)

(main effect of Difficulty: $F(1/14) = 12.6, p < 0.01$). All subjects responded faster in the pattern matching task (1.69 sec) than in the arithmetic task (2.59 sec) (main effect of Task: $F(1/14) = 76.0, p < 0.01$).

3.2. EEG power spectral density analysis

During the S1-interval the activity within the beta range increased in anticipation of the difficult tasks, but decreased in anticipation of the easy tasks from the first to the second interval of analysis within the 6 sec interval. These different tendencies are confirmed by an interaction of Difficulty and Interval ($p < 0.05$, see table 1 for F-values).

Fig. 1 illustrates the laterality effects in beta power. In anticipation of the arithmetic tasks beta activity was more pronounced at C3 than at C4 as was confirmed by the t-test ($t = 3.75, p < 0.01$). There was more left-hemispheric beta activity preceding the arithmetic tasks than preceding pattern presentation ($t = 3.56, p < 0.01$). This development is also confirmed by the interaction Task and Hemispheres (see table 1). No significant differences in beta activity between the two hemispheres were found in anticipation of the pattern tasks. Fig. 1, furthermore, demonstrates that the task-dependent asymmetries become more pronounced during the second 2.56 sec interval within the S1-interval (see table 1; interaction of Task, HE, and Interval, $p < 0.05$).

No significant effects were provided for frequencies within the alpha band.

3.3. Slow cortical potentials

Fig. 2 illustrates slow potential shifts averaged separately for derivations (C3, C4, T3, T4) and conditions (tasks and difficulty). The PCA determines three com-

Table 1
Analysis of variance for repsonse speed, EEG beta power, early and late component of slow cortical potentials with the factors Difficulty (easy vs. difficult), Task (arithmetic vs. patterns), Topography (temporal vs. precentral), Hemispheres (left vs. right), and Interval (first 2.56 sec, second 2.56 sec of S-1 interval). Degrees of freedom: 1/14. Only F-values with $p < 0.05$ are listed.

	RT	Beta EEG	Early SCP	Late SCP
Difficulty	12.6 [a]			
Task	76.0 [a]			
Topography			41.9 [a]	4.9 [b]
Task × HE		5.9 [b]		
Task × Difficulty × HE				5.5 [b]
Difficulty × Int		5.5 [b]		
Task × Int × HE		8.7 [b]		

[a] $p < 0.05$. [b] $p < 0.01$.

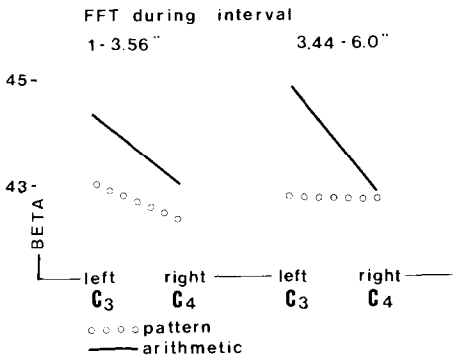

Fig. 1. Beta EEG power of the left and right hemisphere (recording sites: C3 and C4) analyzed separately for arithmetic tasks and pattern matching tasks, and for the first and second half of the S1-interval. Ordinate denotes relative power in percent.

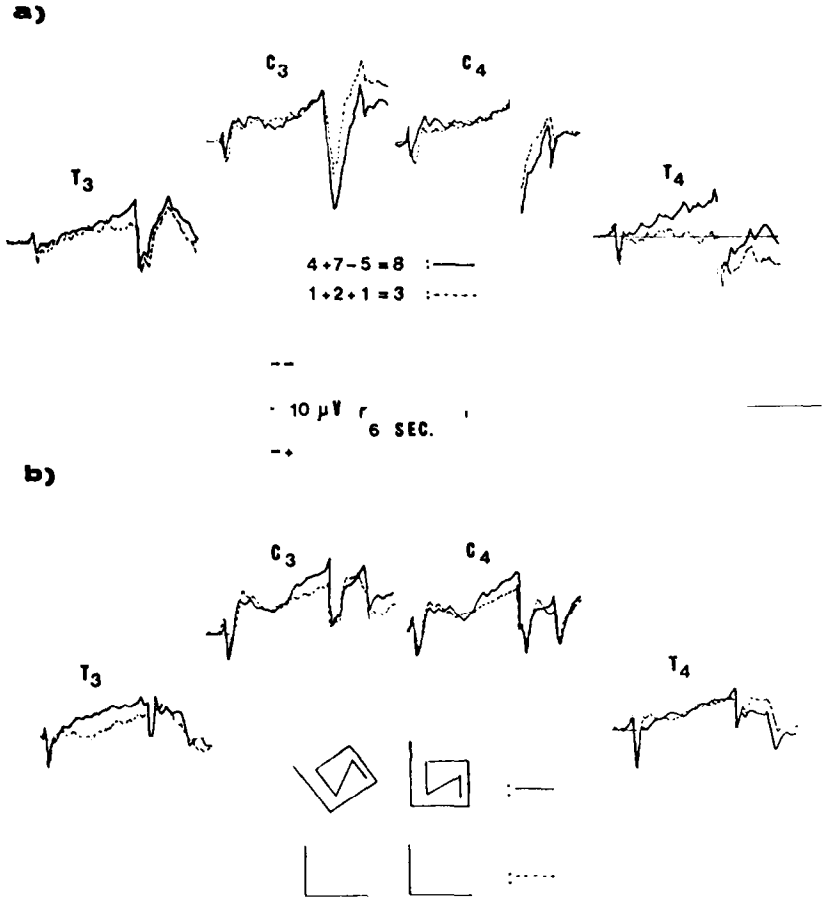

Fig. 2. Slow potential shifts, averaged separately for the recordings (T3, T4, C3, C4), the tasks (a: arithmetic tasks; b: pattern matching tasks), and levels of difficulty (easty tasks: ·····, difficult tasks: ———).

ponents for the S1-interval: an 'early' component with maximum loadings at about 1.5 sec following S1-onset; a second component, labeled 'intermediate', with maximum at about 3 sec following S1-onset; and a third, 'late' component peaking just prior to S2-onset (see fig. 3).

The early component shows pronounced negative amplitudes at C3 and C4, but small positive values at T3 and T4 (see table 1: main effect of Topography, $p < 0.01$). No significant effects are found for the intermediate component.

Fig. 2 illustrates a late negative shift at all recording sites. A significant main effect of Topography ($p < 0.05$, see table 1) demonstrates that the amplitude of the late component is more pronounced precentrally than temporally. This topographical differentiation tends to be less pronounced in anticipation of the difficult task than in anticipation of the easy task, as reflected by a weak interaction of Diff X TO ($F(1,14) = 4.2, p < 0.1$).

Higher negative shifts in anticipation of the difficult tasks as compared to the easy ones were recorded at the temporal recording sites in the present investigation with non-cephalic reference. The late component shows a task-dependent hemispheric asymmetry, but only in anticipation of the easy tasks. The negativity tends to be more pronounced over that hemisphere which is thought to process the expected task. No consistent asymmetry if found prior to difficult tasks. These

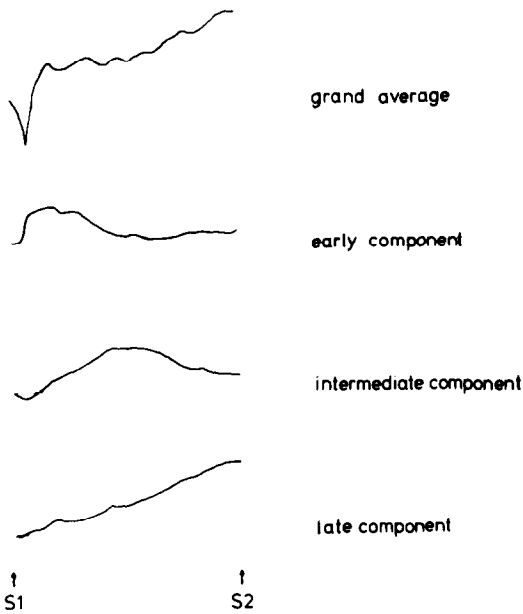

Fig. 3. Grand mean and three components of the SCPs determined by the PCA for the S-1 interval.

tendencies are reflected by the three-way-interaction Diff, Task, and HE (see table 1; if separate ANOVAs are computed for the easy and for the difficult tasks, the two-way interaction Task × HE is significant for the easy tasks, but not for the difficult ones).

4. Discussion

Three components are extracted for the SCPs by a PCA of the present data. This component structure is similar to several other studies (see Lutzenberger et al., this issue): the early component, with a frontal maximum, shows values around zero at the parietal recording. In the present data no early component was observed at temporal leads too. The present data do not indicate any hemispheric asymmetry for the early component.

The level of task difficulty influenced the late negative SCP component and beta activity. While no hemispheric asymmetries were observed in anticipation of the difficult tasks, the late component was more pronounced over the left than over the right hemisphere during the foreperiod of the easy arithmetic task. The anticipation of more difficult tasks may induce increased preparation for processing and processing activity in both hemispheres, so that asymmetries in SCPs may have disappeared. This interpretation receives support from the relatively pronounced late component at T3 and T4 prior to the difficult tasks.

Task-dependent asymmetries of the beta activity turned out to be less influenced by the task difficulty. However, beta activity increased within the anticipation interval, whenever difficult tasks were signalled, but decreased during the anticipation of easy tasks. Although the present results do not allow definitive conclusions, they point at two critical variables that impair the measurement of hemispheric asymmetries.

(a) The level of task difficulty, which in the present study also affected response speed. This influence of task complexity may be one of the possible reasons why hemispheric asymmetries have not always been found, especially if it is taken into account that SCP shifts were rather small in the present study.

(b) The location of the reference electrodes. Temporal activity increases the problem of finding an inactive reference site. A mastoid reference may actually reduce hemispheric differences. The temporal recordings with non-cephalic reference in the present study demonstrated the temporal activity induced by the present tasks.

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